Lesson 7 Teacher's Checklist / Prep Guide

Complete before class

VIDEO GAME CREATION

New Unit: Core Mechanics

Lessons 7-11 form the "Core Mechanics" unit (collision, scoring, win/lose states, simple UI), per the Course Syllabus's milestone table. Level 1 MVP is done as of Lesson 6 - everything from here builds toward the Level 2 Complete milestone at Lesson 11.

Important: A Bugfix Lives In This Lesson

Two engines - GameMaker and 001 Game Creator - need an actual correction to Lesson 6's win-counting code this week, not just an addition. Both used a generic "this instance was destroyed, for any reason" hook to count kills (GameMaker's Destroy event, 001 Game Creator's "Defeated / Destroyed" trigger). That was fine when the only way an enemy could be destroyed was by being shot - but starting this week, enemies can also be removed by escaping off the bottom of the screen, and that must NOT count as a kill. Both engines' Handouts and Keys walk students through moving the win-counter to the actual hit-detection code instead. Read through the GameMaker and 001 Game Creator Keys before class so this doesn't catch you by surprise.

- ☐ Read the GameMaker and 001 Game Creator Keys' Level 1 sections in full before class - both involve editing existing Lesson 6 logic, not just adding new code.

The other six engines (Scratch, Godot, Unity, GDevelop, TIC-80, Unreal) already tied their win-counter directly to the hit-detection code in Lesson 6, so they need no correction - only the new movement/escape-cleanup logic.

Self-Contained Lesson Assets

No new assets this week - enemy movement and cleanup use only code/logic changes.

Continuity From Lesson 6

- ☐ Remind students to open their own [YourName]Game project from Lesson 6, not create a new one.
- ☐ If a student's Lesson 6 project is lost or broken, have a pre-built "Lesson 6 complete" starter available per engine (from the Lesson 6 Teacher's Keys) so they aren't blocked.

Engine Verification

- ☐ Confirm the demo laptop's Lesson 5 and 6 exercises still work for all 8 engines.

- ☐ GameMaker: walk through the Collision-event fix yourself once before class - it's easy to leave the old Destroy-event decrement in place by accident, double-counting kills.
- ☐ 001 Game Creator (Windows demo laptop): confirm the "Left/Entered Camera" trigger reliably fires for an Actor leaving via the bottom of the screen, not just the top (as used for the bullet in Lesson 3) - same trigger, different edge, worth a quick test.
- ☐ Godot / Unity / GameMaker / GDevelop / Unreal: confirm a slow, continuous downward movement plus an off-screen cleanup check both work smoothly for several enemies at once.
- ☐ TIC-80: confirm the enemies table's per-enemy movement and bottom-of-screen cleanup don't interfere with the existing bullet-collision nested loop.

Materials

- ☐ Print or queue digital copies of all 8 Student Handouts (or confirm students can access their engine's version electronically).
- ☐ Have the 8 Teacher's Keys on hand (instructor reference only - do not distribute) in case a student needs a worked example beyond Level 1.
- ☐ Bring: demo laptop, projector cable/adapter, handout copies.

Known Gotchas This Week

- GameMaker / 001 Game Creator: the single most important thing to watch for this week - confirm the win-counter decrement actually moved to the hit-detection code, not left on the generic destroy hook. Test by letting an enemy escape off the bottom unshot and confirming "You Win!" does NOT appear.
- Every engine: enemies moving continuously means testing takes a few seconds longer each time - remind students to be patient watching a test run through instead of assuming something's broken after 1 second.
- A very slow enemy speed is fine and expected this week - there's no lose condition wired up yet (that's Lesson 8), so a fast enemy just means less time to observe and test.